

Linux administration tips and tricks

Master the command line



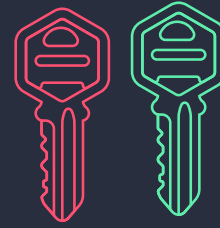
- Get comfortable with the Linux command line.
- The command line your Swiss Army knife for DevOps tasks. Learn basic commands like ls, cd, cp, mv, cat, echo and more.

File Permissions



- Understand file permissions (chmod), ownership (chown), and groups.
- Managing access control is crucial in a DevOps role.

SSH Key Pairs



- Generate SSH key pairs for secure server access. Use ssh-agent to manage keys and never store private keys in plain-text

Firewall Rules



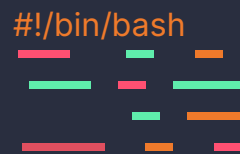
- Learn iptables or use firewalld to control network traffic.
- Properly configuring firewalls is vital for server security.

Take Regular Backups



- Set up automated backups using tools like rsync or tar.
- Regular backups can save your data in emergencies.

Scripting Skills



- Become proficient in Bash or Python scripting to automate repetitive tasks.
- This is the heart of DevOps automation.

Package Management



- Know your package manager (apt, yum, dnf).
- Keep software up to date and resolve dependencies efficiently.

Network Troubleshooting



- Learn iptables or use firewalld to control network traffic.
- Properly configuring firewalls is vital for server security.

Process Management



- Use commands like ps, top, and systemctl to monitor and manage processes.
- Keep an eye on server performance.

Disk Management:



- Master commands like df and du to manage disk space.
- Use LVM for flexible storage management.

Version Control



- Get comfortable with Git for tracking changes to your code and configurations.
- Git is essential for collaboration.

Containerization



- Explore Docker and container orchestration tools like Kubernetes.
- Containers simplify deployment and scaling.

Configuration Management



Learn tools like Ansible, Puppet, or Chef to automate server configuration and ensure consistency.

Cloud Services



- Familiarize yourself with cloud providers like AWS, Azure, or GCP.
- Infrastructure as Code (IAC) is a game-changer.

Testing & CI/CD



- Integrate testing into your DevOps pipelines. Use Jenkins, Travis CI, or GitLab CI for continuous integration and delivery.

Monitoring & Logging



- Implement monitoring with tools like Prometheus, Grafana, or Nagios. Centralized logging (ELK stack) helps troubleshoot issues.

Infrastructure as Code



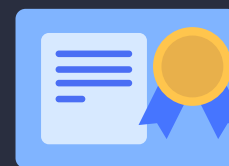
- Write infrastructure code with Terraform or CloudFormation to automate provisioning and scaling of resources.

Automation & Orchestration



- Use tools like Kubernetes, Docker Swarm, or Apache Mesos for container orchestration. Automation streamlines tasks.

Certifications



- Consider certifications like AWS Certified DevOps Engineer, Certified Kubernetes Administrator (CKA), or Red Hat Certified Engineer (RHCE) to validate your skills.

Continuous Learning



- Stay updated with the ever-evolving DevOps landscape. Attend conferences, read blogs, and keep experimenting to excel in your career.